

ZS-A21

Seam-Cohomology Obstruction, Single-Clock Selection, and Polyhedral Graph-Discrimination for ZHCS Dark Matter: Constraining the 32-Face Abundance Degeneracy

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Verification Summary

Verification: **28/28 computational checks PASS** (Appendix I; script released) | Zero new fitted parameters | $(A, Q, \dim Z) = (35/437, 11, 2)$ **LOCKED** | **Theorem 1 (Seam-Cohomology Obstruction): PROVEN** — with the A19 one-cross-edge graph $b_1(\Gamma_{38}) = 66$, $H^3_{\text{seam}} \cong \mathbb{Z} \oplus \mathbb{Z}^{\wedge\{66r\}} \oplus T^{\wedge\{66\}}$; unique parent $\Leftrightarrow H_1(M_Z) = 0$ | **Corollary 2:** single-clock + no-torsion $\Leftrightarrow M_0$; $M_0 =$ **DERIVED-CONDITIONAL on L_{clock}** (reduced here to $L_{\text{clock}} = L_{\text{clock-bulk}} \wedge L_{\text{clock-restrict}}$) | **Uniform Nodewise Coupling Theorem: PROVEN within U1–U6 $\Rightarrow G_Z$ reduced to $g_1 \wedge g_2$** | **Theorem 4 (Polyhedral Graph-Discrimination): PROVEN** — the icosidodecahedron is **incompatible with the A19-fixed seam graph** (its cold $b_1 = 29$ forces seam $b_1 = 36 \neq 66$; not an independent exclusion), and the truncated icosahedron and truncated dodecahedron are **non-isospectral by the exact integer invariant** $\text{tr}(L^2) = \sum d_i^2 + 2E = 1200$ vs 1560; the graph-to-physical-mode bridge is the open lemma **L_{spec} (TARGET)**, so the 32-face abundance degeneracy is constrained at the graph level | Physical assignment: **HYPOTHESIS-strong**, now with a concrete discriminator | Empirical program = **TARGET-COMPUTATIONS** (ILLUSTRATIVE ratio-only selection; NON-CLAIM) | Fit-free: $R_{\text{cb}} = 16/3$, $f_{b^{\wedge\{cb\}}} = \Omega_b/(\Omega_c + \Omega_b) = 3/19$, $\Omega_{\text{cb}} = 38/121$ ($n_{\square} = 0.9674$ imported). **No real-data result is claimed.**

§0. Abstract

ZS-A21 v1.3 corrected the seam Betti number and baryon fraction and scoped the coupling theorem to an explicit ansatz; v1.4–v1.5 keep that corrected base and add a new result that **constrains the 32-face abundance degeneracy at the graph level**, a sharpening of L_{clock} , and status corrections; v1.5 finalizes the framing and the proofs and is the terminal text version, all under claim-to-tool calibration.

(i) Seam-Cohomology Obstruction (Theorem 1, PROVEN). With $H_1(M_Z; \mathbb{Z}) = \mathbb{Z}^r \oplus T$ and $m = b_1(\Gamma)$, Künneth and Poincaré–Lefschetz give $H^3((M_Z, \partial M_Z) \times \Gamma; \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}^{\wedge\{mr\}} \oplus T^{\wedge\{m\}}$. For the A19 one-cross-edge seam graph, $b_1(\Gamma_{38}) = 59 + 7 = 66$; a unique parent charge exists iff $H_1(M_Z) = 0$.

(ii) Homology-Ball Selection (Corollary 2) and a sharper L_{clock} . single-clock + no-torsion $\Leftrightarrow M_0$ (a corollary of Theorem 1 and standard 3-manifold homology). The physics lemma L_{clock} is sharpened into a conjunction: **$L_{\text{clock-bulk}}$** (the A20 scalar kinetic matrix is rank-one, $S_{\text{cb}} = 0$; IMPORTED from A20) and **$L_{\text{clock-restrict}}$** (its restriction to the seam boundary is rank-preserving). M_0 is thus

DERIVED-CONDITIONAL on $L_{\text{clock}} = L_{\text{clock-bulk}} \wedge L_{\text{clock-restrict}}$; we do not claim M0 outright, and the explicit BV–BFV zero-mode computation of $L_{\text{clock-restrict}}$ is named as the principal mathematical-physics target.

(iii) Uniform Nodewise Coupling Theorem (PROVEN within U1–U6). Within an explicit ansatz, graph-edge conservation $d\Gamma k = 0$ on the connected Γ_{38} forces a single coefficient $k_v = k$ and $|k| = 1$ ($k = +1$ by orientation), discharging g_3 and g_4 and reducing G_z to $g_1 \wedge g_2$ (OPEN); the unrestricted coupling classification remains OPEN. The I_h -invariant cold functional is two-dimensional (12-pentagon and 20-hexagon orbits), and $a = b$ is forced by graph connectedness, not I_h .

(iv) Polyhedral Graph-Discrimination Theorem (Theorem 4, PROVEN) — the new result. Three solids realize 32 faces and hence the abundance $32/121$ identically — the truncated icosahedron (12 pentagons + 20 hexagons), the truncated dodecahedron (20 triangles + 12 decagons), and the icosidodecahedron (20 triangles + 12 pentagons) — so a ratio-only likelihood is polyhedron-blind. We constrain the degeneracy at the graph level with two corpus-internal invariants. (a) **Topological compatibility:** the cold-module face graph has $b_1 = 59$ for the first two but 29 for the icosidodecahedron, so the icosidodecahedron is **incompatible with the A19-fixed seam graph** (it would force seam $b_1 = 36 \neq 66$). This is a compatibility statement with A19's already-truncated-icosahedron specification, **not an independent selection theorem** ($b_1 = 66$ is itself computed from the A19 graph; a true selection would require $b_1 = 66$ derived independently — e.g. from an action zero-mode count or from L_{clock}). (b) **Spectral non-isospectrality (exact):** the truncated icosahedron and truncated dodecahedron share $(V, E, b_1) = (32, 90, 59)$ but have different Laplacians, proven without floating point by the integer invariant $\text{tr}(L^2) = \sum d_i^2 + 2E = 12 \cdot 5^2 + 20 \cdot 6^2 + 2 \cdot 90 = 1200$ versus $20 \cdot 3^2 + 12 \cdot 10^2 + 2 \cdot 90 = 1560$. The bridge from the graph spectrum to a physical mode spectrum is a separate open lemma **L_{spec}** (that the linearized ZHCS boundary Hessian contains the cold-module Laplacian with a fixed coefficient, giving $\omega_{\square}^2 = \omega_0^2 + \alpha \lambda_{\square}$); the CMB template is a TARGET-COMPUTATION. The upstream ZS-F2/A19 structure (the 12/20 orbit assignment, the C_{60} fullerene realization, $|I_h| = 120 = Q^2 - 1$) selects the truncated icosahedron among the two. The physical assignment is thereby narrowed — the icosidodecahedron incompatible with A19, the remaining pair exactly distinguishable as graphs — HYPOTHESIS-strong, with the graph-to-physics bridge (L_{spec}) and the template still open.

(v) Empirical program (TARGET + ILLUSTRATIVE). The model ladder (with electron-mass step clarified), discrete-geometry Bayes factors, held-out tests, custom-CLASS/CAMB, and the fit-free observables $R_{\text{cb}} = 16/3$, $f_{\text{b}}^{\{(\text{cb})\}} = 3/19$, $\Omega_{\text{cb}} = 38/121$ ($n_{\square}$ imported separately) are pre-registered; an illustrative ratio-only selection gives $P(16/3 \mid D) \approx 1$ within the discrete set but a weak discrete-versus-continuous Bayes factor ($B \approx 1.8$). The honest terminus: Theorems 1, 4 PROVEN, M0 DERIVED-CONDITIONAL on a sharpened L_{clock} , G_z halved to $g_1 \wedge g_2$, the 32-face degeneracy broken in principle, and the empirical program fully specified with no real-data result asserted. Zero new fitted parameters; **(A, Q, dim Z) = (35/437, 11, 2) LOCKED**.

Epistemic Status Legend

Table 0. Status tags used in this paper (uppercase, bold).

STATUS	DEFINITION
PROVEN	Theorem from standard mathematics or corpus definitions; machine-verifiable.

STATUS	DEFINITION
PROVEN within U1–U6	Theorem holding under the explicitly stated coupling ansatz U1–U6 (not over all couplings).
DERIVED	Follows from the Z-Spin action plus PROVEN inputs; zero free parameters beyond A.
DERIVED-CONDITIONAL	Derived, conditional on named, independently falsifiable conditions ($L_{\text{clock-restrict}}$, g_1 , g_2).
IMPORTED-PROVEN	Theorem proven externally (or upstream, e.g. A20); its Z-Spin use is a separate identification.
LOCKED	Core constant fixed upstream ($A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$); no downstream paper may modify.
TARGET-COMPUTATION	A well-posed computation not yet executed.
ILLUSTRATIVE	A toy/limited-scope computation demonstrating the method; a NON-CLAIM about the real quantity.
HYPOTHESIS-strong	Multiple independent structural anchors; promotion path documented; not yet PROVEN/DERIVED.
OBSERVATION	A statement of consistency with published data, not a fit.
OPEN	A sharply stated, falsifiable question current corpus tools cannot settle.
NON-CLAIM	An explicit declaration of what is not asserted.

§1. Introduction and Advancement over v1.1–v1.3

The chain $ZS-A18 \rightarrow A19 \rightarrow A20 \rightarrow A21$ v1.1 (Global Parent-Charge Theorem under M0) $\rightarrow$ v1.2 (obstruction classification; single-clock selection; partial Gz closure; discrete-geometry program) $\rightarrow$ v1.3 (corrections: $b_1 = 66$, $f_{b^{\{(cb)\}}} = 3/19$, the coupling theorem scoped to U1–U6, the I_h contradiction resolved) reaches its most stable form. The review of v1.3 noted, correctly, that further value comes not from prose but from concrete computations. Acting on that, v1.4 computed a corpus-internal invariant and v1.5 finalizes it: a graph invariant that **constrains the 32-face abundance degeneracy at the graph level** (Theorem 4, now stated as a compatibility-plus-exact-non-isospectrality result), a **sharpened L_{clock}** (two falsifiable sub-lemmas), and the status corrections (the BV–BFV zero-mode wording, the coupling level k , and the electron-mass ladder).

The degeneracy result is the lead. The review observed that three distinct 32-face solids give the same abundance, so a ratio-only likelihood cannot select among them, and asked for a shape-sensitive observable. We go one step further at the graph level: the A19 seam graph requires the cold module to have $b_1 = 59$, with which the icosidodecahedron ($b_1 = 29$) is incompatible — a consistency statement with A19's specification, **not an independent selection** (the value 66 is computed from that graph); and the remaining pair, the truncated icosahedron and the truncated dodecahedron, while sharing (V, E, b_1) , are exactly non-isospectral by $\text{tr}(L^2) = 1200$ versus 1560. The organizing question is unchanged; the new content narrows the physical-assignment residual that v1.3 left fully open, with the graph-to-physics bridge (L_{spec}) named as the remaining step.

§2. Inherited Framework (v1.1–v1.3 recap)

Carried unchanged: the absolute- H^3 no-go on the polyhedral surface; the relative flux $H_Z = \iota_*(\ast J)$ classified by the relative Dixmier–Douady class in $H^3(M_Z, \partial M_Z; \mathbb{Z})$ (a bulk-trivial gerbe with a nontrivial boundary trivialization, pulled back under $\iota: M_Z \rightarrow M_4$); the single harmonic parent mode $H^0(\Gamma_{38}; \mathbb{Z}) = \mathbb{Z}$;

the augmentation functionals $\varepsilon_C(1_{38}) = 32$, $\varepsilon_B(1_{38}) = 6$ ($Q_C : Q_B = 16/3$); and the abundance normalization $\Omega_C^{\square} = 32/121$, $\Omega_B = 6/121$ as the canonical trace $\tau_{121}(P) = \text{Tr } P/121$ on $H_{\text{ch}} \cong \mathbb{C}^{121}$ ($P_C, P_B \in M_{121}(\mathbb{C})$, ranks 32, 6). The locked inputs ($A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$) introduce no parameter beyond A and Q .

§3. The Seam-Cohomology Obstruction Theorem (Theorem 1)

For a connected compact oriented 3-manifold M_Z with boundary and a connected finite graph Γ , Künneth ($H^1(\Gamma)$ free, Tor vanishes) and Poincaré–Lefschetz duality ($H^3(M_Z, \partial M_Z) \cong H_0(M_Z) \cong \mathbb{Z}$, $H^2(M_Z, \partial M_Z) \cong H_1(M_Z)$), with $H_1(M_Z; \mathbb{Z}) = \mathbb{Z}^r \oplus T$ and $m = b_1(\Gamma)$, give

$$H^3((M_Z, \partial M_Z) \times \Gamma; \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}^{mr} \oplus T^{\otimes m}. (1)$$

The A19 seam graph is the truncated-icosahedron face graph ($V = 32$, $E = 90$, $b_1 = 59$), the cube-face graph ($V = 6$, $E = 12$, $b_1 = 7$), and one cross-edge; a bridge creates no cycle, so $b_1(\Gamma_{38}) = 59 + 7 = 66$ ($E = 103$, $V = 38$). **Theorem 1.** With $m = 66$, a unique integer parent charge exists **iff** $H_1(M_Z) = 0$; otherwise each free generator opens 66 continuous channels ($N_{\text{cont}} = 66r$) and each torsion generator a discrete sector ($T^{\wedge \{ \otimes 66 \}}$); the free ranks for $r = 0, 1, 2$ are 1, 67, 133. Status: **PROVEN** (Appendix I). This identifies M_0 as the exact single-parent condition and, as §8 shows, its multiplicity 66 also discriminates among the candidate cold polyhedra.

§4. BV–BFV Zero Modes and the Mode–Cohomology Correspondence

A Hodge decomposition of the relative gerbe field gives the parent part, the obstruction modes of Theorem 1, and an exact remainder; the free continuous-mode count is $N_{\text{cont}} = 66r$ and torsion gives discrete sectors $T^{\wedge \{ \otimes 66 \}}$. Two statuses must be kept separate. In the **abstract** abelian BF / gerbe model, these are exactly the zero modes of the boundary BV–BFV phase space (Cattaneo–Mnev–Reshetikhin): **IMPORTED-PROVEN**. Their identification with the **actual** ZHCS seam modes is a separate step, conditional on the identifications $g_1 \wedge g_2$ of §6: **DERIVED-CONDITIONAL on $g_1 \wedge g_2$** (the v1.2/v1.3 phrasing “these are exactly the zero modes” is here qualified as holding for the abstract model and conditionally for ZHCS).

§5. Homology-Ball Selection (Corollary 2) and a Sharper L_{clock}

Corollary 2 (Homology-Ball Selection). With $\partial M_Z = S^2$ and M_Z connected oriented, the ZHCS seam zero-mode phase space is a single continuous parent clock with no torsion sector **iff** M_Z is an integral homology 3-ball (M_0). This is Theorem 1 combined with the standard fact that $H_1 = 0$, $\partial M = S^2$ forces a homology ball; its value is the physical reading (single-clock dynamics as a topology-selection rule). Status of the equivalence: **PROVEN** (pure-mathematics novelty moderate; mathematical-physics novelty high).

Sharpening L_{clock} . The physics content beyond the equivalence is the lemma L_{clock} — that the actual seam zero-mode phase space is rank-one and torsion-free. We factor it: **$L_{\text{clock-bulk}}$** , that the A20 scalar kinetic matrix is rank-one with $S_{\text{cb}} = 0$ (this is A20's single-clock result, **IMPORTED** from A20), and **$L_{\text{clock-restrict}}$** , that the restriction of that kinetic form to the seam boundary is rank-preserving (no extra boundary zero mode appears and none is lost). Then $L_{\text{clock}} = L_{\text{clock-bulk}} \wedge L_{\text{clock-restrict}}$, and M_0 is **DERIVED-CONDITIONAL on $L_{\text{clock-restrict}}$** (given $L_{\text{clock-bulk}}$ from A20).

$L_{\text{clock-restrict}}$ is exactly a boundary BV–BFV zero-mode computation of the A19/A20 boundary action — a precise, falsifiable statement within [11–13], not performed here, and named as the principal mathematical-physics target. **NON-CLAIM:** M0 is not claimed derived outright.

§6. The Uniform Nodewise Coupling Theorem

With the seam field a coefficient-lattice-valued relative differential character $H_z \in \hat{H}^3(M_4, \partial M_4; \Lambda\Gamma)$ and a degree-one clock character dT , the topological coupling $S_{\text{top}} = 2\pi k \int H_z \cup dT$ reproduces the A20 parent action's topological term. Under the explicit ansatz **U1–U6** (U1 bilinear in H_z and one clock character; U2 degree 3+1, first-order; U3 node-diagonal integer coefficients; U4 no background characteristic class or higher cup operation; U5 graph-edge conservation; U6 boundary counterterms modded out; **U7 k is primitive** (minimal-level sector)):

$d_{\Gamma} k = 0$ on connected $\Gamma_{38} \Rightarrow k_v = k \in \mathbb{Z}$ (U1–U6); $|k| = 1$ only under U7 (primitive), $k = +1$ by orientation.(2)

Theorem 3 (Uniform Nodewise Coupling), PROVEN within U1–U6. The connected-graph kernel is exact, so $k_v = k \in \mathbb{Z}$ under U1–U6; the minimal level $|k| = 1$ holds under U7 (large gerbe-gauge invariance alone gives only $k \in \mathbb{Z}$, and primitivity gives $|k| = 1$). **NON-CLAIM:** uniqueness over all local couplings (boundary counterterms, characteristic-class couplings, torsion operations, off-diagonal pairings, higher cup operations) is OPEN. Within U1–U6, g_4 (single coefficient) and g_3 (integral lattice $\Rightarrow$ the augmentation pairings) are discharged, reducing $Gz = g_1 \wedge g_2 \wedge g_3 \wedge g_4$ to $g_1 \wedge g_2$ (OPEN). C1-ID-global is **DERIVED-CONDITIONAL on M0** ($\leftarrow L_{\text{clock-restrict}}$), on $g_1 \wedge g_2$, and on U1–U6.

§7. The Augmentation: $a = b$ from Connectedness, not I_h

The 32 truncated-icosahedron faces split into two I_h orbits (12 pentagons, 20 hexagons), so the I_h -invariant integer functionals on the cold module are **two-dimensional**, $\varepsilon_{\{a,b\}}(v) = a \cdot \Sigma_{\text{pent}} v_i + b \cdot \Sigma_{\text{hex}} v_j$; I_h alone does not force $a = b$. What forces $a = b = k$ is the full 38-node graph-edge conservation (§6, $d_{\Gamma} k = 0$) with graph connectedness — a single coefficient over all 38 nodes — so $\varepsilon_C(1_{38}) = 12 + 20 = 32$ and $\varepsilon_B(1_{38}) = 6$, giving $Q_C : Q_B = 16/3$. The uniqueness rests on connectedness and U1–U6, tying §7 to Theorem 3. Status: ratio **DERIVED-CONDITIONAL on $g_1 \wedge g_2$ and U1–U6** (mathematics PROVEN under M0).

§8. Constraining the 32-Face Abundance Degeneracy (Theorem 4)

Three solids realize 32 faces and hence the abundance $\Omega_{\text{cdm}} = 32/121$ identically: the truncated icosahedron (12 pentagons + 20 hexagons), the truncated dodecahedron (20 triangles + 12 decagons), and the icosidodecahedron (20 triangles + 12 pentagons). A cosmological likelihood that uses only the abundance ratio is therefore **polyhedron-blind**. Two corpus-internal observables break the degeneracy.

Table 1. The three 32-face solids and their cold-module (face-adjacency) graphs; $\text{tr}(L^2) = \Sigma d_i^2 + 2E$ is exact (Appendix I).

Solid (cold module)	Faces	Face graph (V, E, b_1)	seam b_1	$\text{tr}(L^2)$
truncated icosahedron	12 pent + 20 hex	(32, 90, 59)	66	1200

Solid (cold module)	Faces	Face graph (V, E, b_1)	seam b_1	$\text{tr}(L^2)$
truncated dodecahedron	20 tri + 12 decagon	(32, 90, 59)	66	1560
icosidodecahedron	20 tri + 12 pent	(32, 60, 29)	36 ($\neq 66$)	600

Theorem 4 (Polyhedral Graph-Discrimination). (i) **Topological compatibility:** the cold-module face graph has $b_1 = 59$ for the truncated icosahedron and the truncated dodecahedron but $b_1 = 29$ for the icosidodecahedron; hence the icosidodecahedron is **incompatible with the A19-fixed seam graph**, which has $b_1(\Gamma_{38}) = 66 (= 59 + 7)$ — the icosidodecahedron would force seam $b_1 = 36$. This is a **consistency statement with A19's already-truncated-icosahedron specification, not an independent selection theorem**: the value 66 is itself computed from the A19 graph, so it cannot, on its own, independently select that graph. A genuine selection would require $b_1 = 66$ to be fixed independently of the polyhedron choice (e.g. by an action zero-mode count, by L_{clock} , by an observable mode multiplicity, or by an upstream symmetry theorem). (ii) **Spectral non-isospectrality (exact):** the truncated icosahedron and truncated dodecahedron have identical $(V, E, b_1) = (32, 90, 59)$ face graphs but different Laplacians, proven without floating point by the integer invariant $\text{tr}(L^2) = \sum d_i^2 + 2E$: for the truncated icosahedron (12 pentagons of degree 5, 20 hexagons of degree 6) $\text{tr}(L^2) = 12 \cdot 25 + 20 \cdot 36 + 180 = 1200$, while for the truncated dodecahedron (20 triangles of degree 3, 12 decagons of degree 10) $\text{tr}(L^2) = 20 \cdot 9 + 12 \cdot 100 + 180 = 1560$; $1200 \neq 1560$. Status: (i) **PROVEN** (a compatibility statement, not a selection); (ii) **PROVEN** (exactly non-isospectral; Appendix I).

Physical realization, lemma L_{spec} , and status. The discriminator is a graph invariant; its physical relevance requires a bridge we state as an open lemma. **L_{spec} :** the linearized ZHCS seam quadratic action has the form $S^{(2)} = \frac{1}{2} \int d\eta [\varphi^T K \varphi - \varphi^T (\omega_0^2 I + \alpha L_{\Gamma} c) \varphi]$, so that its boundary Hessian contains the cold-module Laplacian $L_{\Gamma} c$ with a fixed coefficient α ; then the mode frequencies are $\omega_{\square}^2 = \omega_0^2 + \alpha \lambda_{\square}$ for the graph eigenvalues $\lambda_{\square}$, and the truncated-icosahedron and truncated-dodecahedron spectra induce distinct seam-mode spectra and hence distinct shape-sensitive perturbation/isocurvature templates. The explicit status ladder is therefore: graph non-isospectrality **PROVEN** (Theorem 4(ii)); the graph spectrum equals the ZHCS mode spectrum **HYPOTHESIS / TARGET** (this is L_{spec} , requiring the explicit linearized boundary action); the CMB template **TARGET-COMPUTATION**. The rank-ratio likelihood (§10) tests the rank pair $32 : 6$ but is polyhedron-blind. The upstream ZS-F2/A19 structure (the 12/20 pentagon–hexagon orbit assignment, the C_{60} fullerene realization, $|I_h| = 120 = Q^2 - 1$) selects the truncated icosahedron among the two. The physical-assignment residual is thereby narrowed from a three-fold degeneracy that v1.3 left fully open to: the icosidodecahedron **incompatible with the A19 specification**, and the remaining pair **exactly distinguishable as graphs** — still HYPOTHESIS-strong (upstream selects the truncated icosahedron), with the graph-to-physics bridge L_{spec} and the template realization the remaining open steps.

§9. The Equivariant Seam Complex (remark)

Equivariant bundle gerbes and equivariant differential cohomology exist externally and are not novel here. The object whose construction would be new is the total complex $H^3_{\{\text{seam}, G\}}(M_Z, \partial M_Z; \Gamma, \Lambda)$ combining relative differential cohomology, the finite-graph cochain complex, the coefficient lattice, the I_h/O_h action, and BV–BFV gluing; its target properties (subdivision invariance, additivity under seam gluing, naturality under graph refinement, the equivariant augmentation's universal property, functoriality

of the obstruction group, and an unrestricted differential-cohomology coupling classification) are stated as a **CONJECTURE** within the BV–BFV/cellular-BF frameworks. Status: **OPEN** (a documented extension; the pure-mathematics 9-point target). It is flagged so the paper is not read as asserting a gluing theorem it has not proven.

§10. The Discrete-Geometry Likelihood Program (folded ZS-A22 scope)

The empirical paper is folded here to limit paper proliferation; every execution is declared TARGET. The model ladder isolates which lock drives any result, with sampled parameters specified per row. The electron-mass step is clarified: $m_e(z)/m_e(0) = 1$ is the standard recombination value, so to make the lock non-vacuous M_N **fixes** the electron-mass ratio as a sampled parameter and M_Z **fixes** it to 1 (M_Z is then a proper nested restriction of M_N , testing whether ZS-C's prediction of no electron-mass variation is preferred). In M_R only the ratio is fixed, so one absolute-density parameter $\omega_cb = \omega_c + \omega_b$ is sampled, with $\omega_b = (3/19)\omega_cb$, $\omega_c = (16/19)\omega_cb$.

Table 2. The pre-registered model ladder (each compared to the public Planck 2018 Λ CDM baseline).

Model	Fixed condition	Sampled cosmological parameters
M_Λ	none (baseline)	$\omega_b, \omega_cdm, \theta_s, \tau, A_\square, n_\square$ (standard six)
M_R	$\omega_c/\omega_b = 16/3$	$\omega_cb, \theta_s, \tau, A_\square, n_\square$
M_F	$\Omega_b = 6/121, \Omega_c = 32/121$	$h, \tau, A_\square, n_\square$
M_N	$M_F + n_\square = 0.9674$; electron mass free	$h, \tau, A_\square, m_e(z)/m_e(0)$
M_Z	full ZS-C ($M_N + m_e(z)/m_e(0) = 1$)	$h, \tau, A_\square$ (residual only)

This yields the separated increments $\Delta\chi^2_R, \Delta\chi^2_F, \Delta\chi^2_N, \Delta\chi^2_Z$ and, by the $M_N \rightarrow M_Z$ step, an isolated test of the electron-mass branch. The discrete-geometry test reframes the anti-numerology question (§14) on the data side: pre-register the candidate ratio set R_geom from the admissible polyhedra (symmetry group, sector dimension, orbit structure, rank limits, degeneracy handling, and prior weight all fixed before unblinding), run a restricted cosmology at each ratio r , and compute

$$P(r | D) = Z_r \pi_r / \sum_r Z_r \pi_r, \quad B_{geom,\Lambda} = Z_{disc} / Z_{\Lambda CDM}, \quad B_{geom,cont} = Z_{disc} / Z_{cont}. \quad (3)$$

Illustrative ratio-only result (NON-CLAIM). Treating Planck $\Omega_c/\Omega_b = 5.36 \pm 0.065$ as a single Gaussian datum over the 16 distinct natural polyhedral ratios (equal priors) against a continuous-ratio model (flat prior on $[3, 8]$): within the discrete set $P(16/3 | D) \approx 1.000$ (the next ratio is $\approx 14\sigma$ away), but the discrete-versus-continuous Bayes factor is only $B \approx 1.8$ (weak), and the 32-face count alone is degenerate. **ILLUSTRATIVE only** — the rank-ratio likelihood tests 32 : 6 but, by Theorem 4, is polyhedron-blind; the polyhedron identity needs the seam-mode spectral template. The real Planck/ACT/DESI Bayes factors of (3) and the spectral-template test are **TARGET-COMPUTATIONS**; no observational support is claimed.

§11. Held-Out Prediction and Fit-Free Observables

The fit-free outputs that follow exactly from the charge theorem are the three rationals

$$R_{cb} = \Omega_c/\Omega_b = 16/3, \quad f_b^{(cb)} = \Omega_b/(\Omega_c + \Omega_b) = 6/38 = 3/19, \quad \Omega_{cb} = 38/121. \quad (4)$$

The spectral tilt $n_s = 0.9674$ is **separate**: not a rational, not from the charge theorem, but an upstream prediction imported from ZS-U1. It is essential to write Ω_{cb} , **not** Ω_m : a massive-neutrino contribution makes the total matter density depart from 38/121. For predictive power we pre-register a calibration/validation split: estimate $\{h, \tau, A_s\}$ on $D_{\text{train}} = \text{Planck TT} + \text{lowE}$, freeze, and predict $D_{\text{test}} = \{\text{Planck TE/EE}, \text{Planck lensing}, \text{ACT DR6}, \text{DESI DR2 BAO}, \text{BBN } \omega_b\}$, reporting per-set $\Delta\chi^2_{\text{pred}}$, ELPD, and posterior-predictive p-values. BBN constrains ω_b independently of the CMB; cluster gas-fraction tests of $f_b^{\{cb\}}$ carry large feedback/depletion systematics and serve only as secondary robustness. The held-out evaluations are **TARGET-COMPUTATIONS**; the present comparison of (4) to published central values ($\Omega_c h^2 / \Omega_b h^2 \approx 5.36$) is an **OBSERVATION** of consistency, not a fit.

§12. Custom-CLASS and Cross-Code Verification

The continuum dust is term-by-term the Ma–Bertschinger CDM system (a fluid integrator reproduces standard CDM to machine precision while controls diverge; the full generalized-dark-matter control with the entropy term $-3\mathcal{H}c^2\delta$ deviates 7.4% at $\eta = 50$, Appendix I). The closure replaces the cdm species in CLASS by an independent zhcs species ($\Omega_c^d = 0$, $\Omega_z^c = 32/121$) and cross-checks against unmodified CLASS and CAMB at the same restricted parameters, requiring $\max_{\ell} |C_{\ell}^{\{ZHCS\}} - C_{\ell}^{\{CDM\}}|$ below tolerance. Gates B0–B8 use the numerically stabilized norms. Passing yields VERIFIED-IN-FULL-BOLTZMANN; the compiled run is a **TARGET-COMPUTATION** and a confirmation-of-implementation, not a new observable.

§13. Falsification Gates

Table 3. Multilayer falsification gates for ZS-A21 v1.4.

Gate	Trigger (immediate rejection)
G-A21.1	Γ_{38} disconnected, or its edge set differs from A19's one-cross-edge construction so that $b_1 \neq 66 \Rightarrow$ the obstruction count and Theorem 1 change.
G-A21.2	$L_{\text{clock-restrict}}$ false (the boundary restriction of the A20 kinetic form is not rank-preserving) $\Rightarrow$ M0 is not selected; v1.1's conditional theorem holds but M0 reverts to an assumption.
G-A21.3	A block-dependent topological coefficient survives within U1–U6 ($d\Gamma_k \neq 0$ admissible) $\Rightarrow$ Theorem 3 and the 32 : 6 split fail.
G-A21.4	g_1 (compact gerbe connection) or g_2 (relative trivialization) fails $\Rightarrow$ flux quantization / relative class fails.
G-A21.5	(§8) L_{spec} fails (the linearized ZHCS boundary Hessian does not contain the cold-module Laplacian), or the data prefer the icosidodecahedron (incompatible with the A19 seam graph, seam $b_1 = 36 \neq 66$), or the truncated-dodecahedron $\text{tr}(L^2) = 1560$ template fits better than the truncated-icosahedron $\text{tr}(L^2) = 1200 \Rightarrow$ the graph-discrimination program (Theorem 4) fails.
G-A21.6	Any of B1–B5 fails, or CLASS–CAMB disagree beyond tolerance $\Rightarrow$ implementation or continuum dictionary fails.
G-A21.7	(§10–11) Model ladder shows $p_{\text{restrict}} < 0.05$ for M_F , or a repeated $> 3\sigma$ held-out residual in independent ACT/DESI/BBN $\Rightarrow$ the abundance lock is rejected.

§14. Anti-Numerology

The theorems require no Monte Carlo: H^3_{seam} with $m = 66$ (Theorem 1), the connected-graph kernel of dimension $b_0 = 1$ (Theorem 3), the augmentation values 32, 6, and the polyhedron b_1 and spectra (Theorem 4) are exact. The audited freedom is the physical polyhedron assignment; Theorem 4 now narrows it (the icosidodecahedron is incompatible with the A19 seam graph; the remaining pair is exactly non-isospectral, $\text{tr}(L^2) = 1200$ vs 1560), so the residual freedom is smaller than in v1.3. The enumeration over the natural face-count menu (63 cold×baryon pairs, 16 distinct ratios) shows the Planck-consistent ratio is uniquely 16/3; the rank **values** 32, 6 are PROVEN, the **assignment** HYPOTHESIS-strong. The discrete-geometry program (§10) plus the spectral template (§8) is the data-side completion. The released script reproduces the enumeration, the illustrative selection, and the polyhedron spectra.

§15. Open Residuals and Conclusion

Table 4. Honest open residuals carried by ZS-A21 v1.4 (declared, not closed).

Residual	Statement and path
$L_{\text{clock-restrict}}$	That the A20 kinetic form restricts rank-preservingly to the seam boundary; with $L_{\text{clock-bulk}}$ (IMPORTED from A20) it gives M0 (Corollary 2). Needs a BV–BFV zero-mode computation of the A19/A20 boundary action. OPEN — the principal target.
$g_1 \wedge g_2$	The two surviving G_z identifications (compact gerbe connection; relative trivialization). OPEN (reduced from four within U1–U6).
Unrestricted coupling classification	Whether boundary counterterms, characteristic-class couplings, torsion operations, off-diagonal pairings, or higher cup operations evade U1–U6. OPEN.
Physical assignment (TI vs TD)	The icosidodecahedron is incompatible with the A19-fixed seam graph (Theorem 4(i); a compatibility statement, not an independent exclusion); the truncated icosahedron vs truncated dodecahedron is exactly distinguished as graphs by $\text{tr}(L^2) = 1200$ vs 1560. HYPOTHESIS-strong (upstream selects the truncated icosahedron).
L_{spec} (graph $\rightarrow$ physics)	That the linearized ZHCS boundary Hessian contains the cold-module Laplacian $L_{\Gamma c}$ with a fixed coefficient ($\omega \square^2 = \omega_0^2 + \alpha \lambda \square$), making the graph spectrum a physical mode spectrum and a CMB template. HYPOTHESIS / TARGET — the bridge that would make Theorem 4(ii) observational.
Equivariant total complex	Subdivision/gluing/functoriality and the unrestricted classification (§9); CONJECTURE, the pure-mathematics 9-point target.
Empirical runs	Model ladder, discrete-geometry Bayes factors, held-out predictions, custom-CLASS/CAMB, the spectral-template test. TARGET-COMPUTATIONS; no real-data result claimed.

ZS-A21 v1.5 keeps the corrected, stable base of v1.3–v1.4 and tightens the new result to exactly what is proven. Theorem 1 classifies the seam obstruction ($m = 66$); Corollary 2 reads single-clock dynamics as the topology-selection rule giving M0, DERIVED-CONDITIONAL on the sharpened $L_{\text{clock}} = L_{\text{clock-bulk}} \wedge L_{\text{clock-restrict}}$; the Uniform Nodewise Coupling Theorem discharges g_3, g_4 within U1–U6 (with $k \in \mathbb{Z}$ under U1–U6 and $|k| = 1$ under U7), halving the open conjunction to $g_1 \wedge g_2$; and the **Polyhedral Graph-Discrimination Theorem (Theorem 4)** constrains the 32-face abundance degeneracy at the graph level — the icosidodecahedron is **incompatible with the A19-fixed seam graph** (a compatibility statement, not an independent selection), and the truncated icosahedron and truncated dodecahedron are **exactly non-isospectral** by the integer invariant $\text{tr}(L^2) = 1200$ versus 1560 — with the bridge to a physical mode spectrum named as the open lemma L_{spec} . The ledger is a stable fixed point: each theorem is PROVEN or PROVEN-within-a-stated-ansatz, each physical bridge is a named falsifiable condition ($L_{\text{clock-restrict}}, L_{\text{spec}}, g_1, g_2$), and every computational debt is declared. **This is the**

terminal text version of ZS-A21: the remaining value is gated entirely on the actual computations — a BV–BFV zero-mode proof of $L_{\text{clock-restrict}}$, the explicit linearized boundary action proving L_{spec} , the action-level $g_1 \wedge g_2$, the compiled custom-CLASS/CAMB, and the Planck/ACT/DESI/BBN likelihood — none of which is a text revision. Two minor design refinements are deferred to those computational releases: a 2×2 factorial electron-mass ladder (separating the $n_{\square}$ lock from the electron-mass freedom) and a three-way split of $L_{\text{clock-bulk}}$ into a cosmological-rank statement and a cosmology-to-BFV dictionary. Zero new fitted parameters; $(\mathbf{A}, \mathbf{Q}, \dim Z) = (35/437, 11, 2)$ **LOCKED**.

Acknowledgements & Code Availability

This paper consolidates internal Z-Spin Collaboration deep-exploration notes following ZS-A21 v1.3 and incorporates an internal review that affirmed v1.3 as the stable base and located the next value in concrete computation. Acting on that, v1.4–v1.5 contribute the Polyhedral Graph-Discrimination Theorem (Theorem 4) and a sharpened L_{clock} . The 28 computational checks of Appendix I are reproduced by the released script `zs_a21_verify.py` (NumPy/SciPy): the one-cross-edge 38-node graph ($b_1 = 66$); relative cohomology under M_0 ; the Seam-Obstruction free ranks 1, 67, 133 and torsion decomposition; the connected-graph gradient kernel; the I_h two-orbit augmentation; the 121-dimensional trace; the full-GDM fluid isomorphism; the illustrative ratio-only selection; and the three 32-face polyhedra with their cold-module face graphs ($b_1 = 59, 59, 29$) and Laplacian spectra (truncated icosahedron versus truncated dodecahedron exactly non-isospectral, $\text{tr}(L^2) = 1200$ vs 1560). Only this verification script is released now; the compiled custom-CLASS/CAMB pipeline, the `clik/Cobaya` likelihood, the held-out tests, and the seam-mode spectral template are **future** releases tied to their TARGET runs. This work used AI tools (Anthropic Claude) for verification and drafting; the author assumes full responsibility.

Appendix

A. Relative homology and M_0

For $(\Delta^3, \partial\Delta^3)$: chain ranks (4,6,4,1), boundary through dimension 2, so $H_3(\Delta^3, \partial\Delta^3; \mathbb{Z}) = \mathbb{Z}$, $H_{\leq 2} = 0$; dually $H^3(B^3, S^2; \mathbb{Z}) = \mathbb{Z}$, $H^3(B^3; \mathbb{Z}) = 0$. Under M_0 the same holds for M_z , with $H^2(M_z, \partial M_z; \mathbb{Z}) \cong H_1(M_z; \mathbb{Z}) = 0$.

B. The 38-node seam graph ($b_1 = 66$)

Truncated-icosahedron vertices are even (cyclic) sign-permutations of $(0, \pm 1, \pm 3\phi)$, $(\pm 1, \pm(2+\phi), \pm 2\phi)$, $(\pm\phi, \pm 2, \pm(2\phi+1))$; coplanar facet merging gives 12 pentagons + 20 hexagons (32 faces, 90 edges, $\chi = 2$), face graph connected with $b_1 = 59$. The cube-face graph has 6 nodes, 12 edges, $b_1 = 7$. A19's single cross-edge (a bridge) gives the connected 38-node graph: $E = 103$, Laplacian rank 37, $b_0 = 1$, $\mathbf{b}_1 = 66 = 59 + 7$ — the obstruction multiplicity m . (The coupling-uniqueness kernel of Theorem 3 is the gradient kernel dimension $b_0 = 1$, independent of b_1 .)

C. Künneth and the obstruction (Theorem 1)

$H^3((M_z, \partial M_z) \times \Gamma) = H^3(M_z, \partial M_z) \otimes H^0(\Gamma) \oplus H^2(M_z, \partial M_z) \otimes H^1(\Gamma)$ (Tor vanishes, $H^1(\Gamma)$ free). With $H^3(M_z, \partial M_z) = \mathbb{Z}$, $H^2(M_z, \partial M_z) = H_1(M_z) = \mathbb{Z}^r \oplus T$, $H^1(\Gamma_{38}) = \mathbb{Z}^{\wedge\{66\}}$: $H^3_{\text{seam}} = \mathbb{Z} \oplus \mathbb{Z}^{\wedge\{66r\}} \oplus T^{\wedge\{66\}}$. Free ranks at $r = 0, 1, 2 \rightarrow 1, 67, 133$.

D. BV–BFV zero modes and L_{clock}

In the abstract abelian BF/gerbe model the boundary BV–BFV zero modes are the relative-cohomology classes (IMPORTED-PROVEN); the continuous-mode count $66r$ and discrete torsion sectors follow from §4. $L_{\text{clock}} = L_{\text{clock-bulk}}$ (A20 rank-one kinetic matrix, IMPORTED) $\wedge L_{\text{clock-restrict}}$ (rank-preserving boundary restriction); the latter is a precise BV–BFV zero-mode statement within [11–13], not proven here, and is the principal open target.

E. Differential-character coupling (Theorem 3, within U1–U6)

Under U1–U6: $S_{\text{top}} = 2\pi k \int \text{Hz} \cup dT$; large-gauge invariance quantizes $|k| = 1$ ($k = +1$ by orientation), $d\Gamma k = 0$ on connected Γ_{38} forces $k_v = k$. The unrestricted classification is OPEN; g_3, g_4 follow within the ansatz.

F. Custom-CLASS species

Newtonian: $\delta' = -\theta + 3\Phi'$, $\theta' = -\mathcal{H}\theta + k^2\Psi$; synchronous: $\delta' = -\theta - \frac{1}{2}h'$, $\theta' = -\mathcal{H}\theta$, with $\theta_Z \square c \square = 0$ after cdm removal. Gates B0–B8 with stabilized norms; the future release bundle will include the CLASS fork, the CAMB cross-check, YAML, and git commit.

G. Discrete-geometry illustrative selection

Data $\Omega_c/\Omega_b = 5.36 \pm 0.065$ (Gaussian, illustrative). The 16 distinct natural ratios $\{12,20,32,62\}/\{6,8,14,26\}$; only $32/6 = 5.333$ lies within 1σ (next nearest 4.43, $\approx 14\sigma$). Equal-prior posterior $P(16/3|D) \approx 1.000$; continuous-ratio Bayes factor $B \approx 1.8$ (weak). Real Cl Bayes factors are TARGET; by Theorem 4 the ratio is polyhedron-blind.

H. Polyhedral graph-discrimination (Theorem 4)

The three 32-face solids, built from canonical coordinates and convex-hull face extraction: truncated icosahedron (60 vertices; 12 pentagons + 20 hexagons), truncated dodecahedron (60 vertices; 20 triangles + 12 decagons), icosidodecahedron (30 vertices; 20 triangles + 12 pentagons). Their cold-module face-adjacency graphs have $(V, E, b_1) = (32, 90, 59), (32, 90, 59), (32, 60, 29)$; seam $b_1 = b_1 + 7 = 66, 66, 36$, so the icosidodecahedron is incompatible with the A19-fixed seam graph ($b_1 = 36 \neq 66$). For the remaining pair, non-isospectrality is exact via $\text{tr}(L^2) = \sum d_i^2 + 2E$ (since $\text{tr}(D A) = 0$ and $\text{tr}(A^2) = 2E$): truncated icosahedron $12 \cdot 5^2 + 20 \cdot 6^2 + 2 \cdot 90 = 1200$, truncated dodecahedron $20 \cdot 3^2 + 12 \cdot 10^2 + 2 \cdot 90 = 1560$ (icosidodecahedron $20 \cdot 3^2 + 12 \cdot 5^2 + 2 \cdot 60 = 600$); $1200 \neq 1560$, no floating point. The numerically computed λ_{max} (8.39 versus 12.29) is a consistency cross-check.

I. Verification ledger (28/28)

(A) topology, one-cross-edge graph $b_1 = 66$; (B) relative cohomology under M0; (C) Künneth single parent; (D) augmentation $32/6$; (E) 121-dim trace; (F) full-GDM fluid (perfect dust $\equiv$ CDM; imperfect 7.4%); (H) Seam-Obstruction free ranks $1/67/133$ and $T^{\wedge\{66\}}$; (I) connected-graph kernel $= b_0 = 1$, $|k| = 1$, the I_h two-orbit functional and $a = b$ from connectedness; (J) discrete-geometry toy $P(16/3) \approx 1$, $B \approx 1.8$; (K) polyhedron-identity — 32 faces all (ratio-blind), seam $b_1 = 66/66/36$ (icosidodecahedron excluded), truncated icosahedron versus truncated dodecahedron non-isospectral; and the fit-free observables $16/3, 3/19, 38/121$. Total 28/28 PASS.

References

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- [2] K. Kang, ZS-A18 v1.5 (2026): massless-Goldstone recombination-CDM no-go.
- [3] K. Kang, ZS-A19 v3.1 (2026): ZHCS boundary-tension dust; $\Omega_{\text{cdm}} = 32/121$; 38-node seam graph (one cross-edge).
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Version History

v1.5 (June 2026): Terminal text version — honesty/precision finalization of v1.4 (no new physics program). (1) Theorem 4 renamed **Polyhedral Graph-Discrimination** (from “Polyhedron-Identity”, which over-stated physical identity). (2) Theorem 4(i) reframed as a **compatibility statement** with the A19-fixed seam graph (the icosidodecahedron is incompatible, seam $b_1 = 36 \neq 66$), **not an independent exclusion**, since $b_1 = 66$ is itself computed from that graph. (3) Theorem 4(ii) non-isospectrality made **exact** via $\text{tr}(L^2) = \sum d_i^2 + 2E = 1200$ (truncated icosahedron) vs 1560 (truncated dodecahedron), removing floating-point dependence. (4) The graph→physics bridge named as the open lemma **L_{spec}** (the boundary Hessian contains the cold-module Laplacian; $\omega \square^2 = \omega \square^2 + \alpha \lambda \square$), with the status ladder graph-non-isospectrality **PROVEN** / graph=mode-spectrum **HYPOTHESIS-TARGET** / CMB-template **TARGET**. (5) Theorem 3 level split: $k \in \mathbb{Z}$ under U1–U6; $|k| = 1$ under the added primitivity assumption U7. Deferred minor refinements: a 2×2 factorial electron-mass ladder and a three-way split of L_{clock-bulk}. Verification 28/28 (the spectral check is now the exact integer invariant). Zero new fitted parameters; (A, Q, dim Z) = (35/437, 11, 2) **LOCKED**.

v1.4 (June 2026): Value-leading extension of the corrected v1.3 base. New: **Theorem 4 (introduced as Polyhedron-Identity Discrimination; renamed Polyhedral Graph-Discrimination in v1.5)** constrains the 32-face abundance degeneracy — the three 32-face solids (truncated icosahedron, truncated dodecahedron, icosidodecahedron) give identical 32/121, but the cold-module face graph has $b_1 = 59, 59, 29$, so the icosidodecahedron is incompatible with the A19 seam graph, and the remaining pair, though sharing $(V, E, b_1) = (32, 90, 59)$, is non-isospectral (made exact in v1.5: $\text{tr}(L^2) = 1200$ vs 1560), giving a

shape-sensitive graph discriminator (physical bridge L_{spec} , template realization TARGET). L_{clock} is sharpened into $L_{\text{clock-bulk}}$ (IMPORTED from A20) $\wedge$ $L_{\text{clock-restrict}}$ (a rank-preserving boundary-restriction BV–BFV statement, the principal target). Corrections: §4 split into abstract BF/gerbe IMPORTED-PROVEN versus ZHCS DERIVED-CONDITIONAL on $g_1 \wedge g_2$; the model ladder's electron-mass step clarified (M_N frees $m_e(z)/m_e(0)$, M_Z fixes it to 1, a proper nested restriction); the physical-assignment residual rewritten per Theorem 4. Verification 28/28. Zero new fitted parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.

v1.3 (June 2026): Error-correction of v1.2 — $b_1(\Gamma_{38}) = 66$ (one cross-edge; ranks 1,67,133 and $T^{\{\oplus 66\}}$); baryon fraction $f_b^{\{\text{cb}\}} = 3/19$ (19/3 was the inverse loading), $n \square$ separated as imported; Corollary 2 (renamed from Theorem 2); Uniform Nodewise Coupling Theorem PROVEN within U1–U6 with $|k| = 1$; §7 contradiction resolved ($a = b$ from connectedness, not I_h); model-ladder sampled-parameter column; release phrasing corrected. Verification 25/25.

v1.2 (June 2026): Seam-Cohomology Obstruction Theorem; single-clock selection of M_0 ; action-level partial closure of G_z ; discrete-geometry program with fit-free observables and an illustrative ratio-only selection. (Superseded numerics: $b_1 = 67$, $f_b = 19/3$.)

v1.1 (June 2026): Assumption M_0 (theorem PROVEN under M_0); coefficient-lattice augmentation; 121-dimensional channel notation; bulk-trivial gerbe with boundary trivialization and pullback; G_z as a four-condition conjunction; full-GDM fluid (7.4%).